

Advancing Cybersickness Prediction in Immersive Virtual Reality Using Pre-Trained Large Foundation Models

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ABSTRACT

Cybersickness is one of the problems that users often encounter during virtual reality (VR) experiences. Existing machine learning (ML) and deep learning (DL) methods for predicting cybersickness require massive amounts of high-quality data for effective training, extended training times to achieve accurate predictions, and lack of transferability capability to new VR environments. To address this, we propose a novel approach using zero-shot and few-shot learning mechanisms to leverage the knowledge of pre-trained large foundation models, namely a time series generative pre-trained transformer (TimeGPT) and Chronos. Validated on two open-source VR cybersickness datasets, namely Simulations 2021 and APAL Head 2019 datasets, our fine-tuned TimeGPT model outperforms traditional DL models, achieving superior accuracy (RMSE of 0.26 and 2.69 for the Simulations 2021 and APAL Head 2019 datasets) and significantly reduced training times (up to 81% and 79% for the same datasets) compared to the trained-from-scratch Transformer models.

Index Terms: Human-centered computing—Human Computer Interaction—Virtual Reality—Systems for interaction design

1 INTRODUCTION

Despite the immense popularity of virtual reality (VR), cybersickness remains a significant barrier, causing discomfort such as nausea and headaches that can persist for hours. Several methods exist in the literature on predicting cybersickness, including those that use pre/post-immersive questionnaires and those based on machine learning (ML)/deep learning (DL) methods. Despite the great success of these ML/DL models in predicting cybersickness, they have a few limitations: (i) state-of-the-art (SOTA) ML/DL-based cybersickness prediction models often require higher computational resources and extended training times due to training the model from scratch to achieve accurate predictions [5], (ii) these models require massive amounts of high-quality data for effective training [5], and (iii) these models often lack transferability, limiting their applicability to familiar environments and impeding their performance in unfamiliar environments [5]. Limited research has been conducted to address these above-mentioned research gaps. Researchers have recently explored transfer learning and early-shaping methods, but these approaches still rely on extensive training and large labeled datasets, making them less effective for diverse environments. Recently, pre-trained large foundation models have been widely successful in natural language processing (NLP) and computer vision and offer promising cross-modality knowledge transfer capabilities. However, their potential for VR cybersickness prediction remains unexplored.

This paper proposes an innovative cybersickness prediction method by leveraging the knowledge of large pre-trained foundation models, namely a time series generative pre-trained transformer (TimeGPT) and Chronos, for cybersickness prediction using zero-shot learning (ZSL) and few-shot learning (FSL) without training a

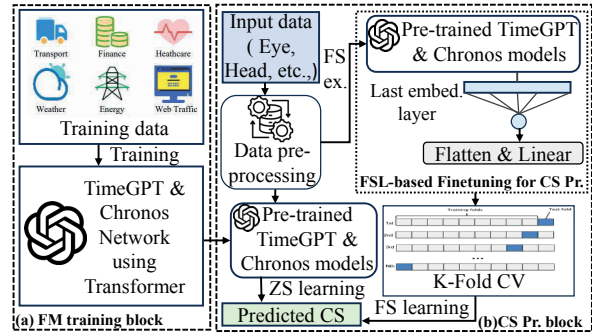


Figure 1: Our proposed ZSL and FSL-based cybersickness (CS) prediction (Pr.) approach (a) Training strategy of foundation models (FM) (e.g., TimeGPT and Chronos), and (b) Use pre-trained TimeGPT and Chronos models as a ZSL inference and fine-tune the TimeGPT and Chronos models using FSL for cybersickness prediction task.

model from scratch. We validate our proposed approach using two open-source VR cybersickness datasets, Simulations 2021 [1], and APAL Head 2019 [3] datasets and compare our proposed method with the traditional cybersickness prediction method (training model from scratch) using six DL-based cybersickness prediction models, namely long short-term memory (LSTM), gated recurrent unit (GRU), convolutional neural network (CNN)-LSTM, neural basis expansion analysis for interpretable time series (N-BEATS), neural hierarchical interpolation for time series (N-HITS), and Transformer model, which are trained with the same datasets from scratch. Our results show that the FSL-based fine-tuned TimeGPT model outperforms traditional DL methods, achieving lower RMSE values (0.26 and 2.69 for the Simulations 2021 and APAL Head 2019 datasets) and 81% and 79% reduction in training time compared to the traditional Transformer model for the same datasets.

2 PROPOSED METHODOLOGY

This section presents the details of our proposed methodology for the cybersickness prediction task by leveraging the knowledge of a large pre-trained FM without training a model from scratch.

Cybersickness Prediction Model Development without FSL and ZSL: At first, for cybersickness prediction, we employ a statistical model such as autoregressive integrated moving average (ARIMA). Then, we develop six DL-based cybersickness prediction models: LSTM, GRU, CNN-LSTM, N-BEATS, N-HITS, and Transformer model for the cybersickness prediction task, which are trained from scratch. We use the Adam optimizer to optimize our DL models with epochs of 200 and a batch size of 256. For training the DL models, we use a learning rate of 0.001. We deploy an early-stopping strategy with a patience value of 30 while training the cybersickness DL models to prevent them from overfitting.

Intelligent Cybersickness Prediction Method Development with FSL and ZSL: In this work, we use two pre-trained large foundation models, namely TimeGPT and Chronos models, by applying ZSL

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and FSL mechanisms to build an intelligent cybersickness prediction method. TimeGPT [4] is a Transformer-based time series architecture with attention mechanisms, which takes a window of historical values to generate the prediction, adding local positional encoding to enrich the input. In contrast, Chronos [2] is a pre-trained probabilistic time series model based on T5 architectures to forecast future patterns. TimeGPT and Chronos models are trained on many publicly available time series datasets with billion data points, including finance, healthcare, etc.. To utilize the TimeGPT and Chronos models for cybersickness prediction tasks, we consider two learning strategies: ZSL and FSL strategy. For ZSL, we use the pre-trained TimeGPT and Chronos model and feed the historical cybersickness measurement data to predict cybersickness severity without adjusting TimeGPT and Chronos model parameters. On the other hand, for FSL, we fine-tune the weights of TimeGPT and Chronos models, starting from their original weights, using domain-specific knowledge (e.g., cybersickness data). To fine-tune the TimeGPT model using FSL, we use the TimeGPT API with a 50 number of iterations, also known as fine-tune steps, to update the weights of the TimeGPT model. In contrast, the Chronos model is fine-tuned with a learning rate of 0.001 over 100 iterations. We validate our proposed method using two open-source datasets: Simulations 2021 [1] and APAL Head 2019 [3]. The Simulations 2021 contain eye tracking, head tracking, and physiological data from 30 participants across five VR simulations, with the Fast Motion Sickness Scale (FMS) score (0–10) serving as the ground truth for discomfort levels. In contrast, the APAL Head 2019 contains head movement data (e.g., head disposition of X, Y, and Z axes, pitch, roll, yaw) from 79 participants in a VR racing game and using the Simulator Sickness Questionnaire (SSQ) as the ground truth for cybersickness severity.

3 EXPERIMENTAL RESULTS

Table 1 shows the mean absolute error (MAE), root mean square error (RMSE), and mean absolute percentage error (MAPE) values for the cybersickness prediction task of the TimeGPT and Chronos foundation models for both with and without fine-tuning methods for the simulations 2021 and APAL Head 2019 datasets. We observe that fine-tuning the pre-trained TimeGPT and Chronos foundation models enhances the cybersickness prediction for both datasets. For instance, the fine-tuned TimeGPT model achieves an RMSE value of 0.26 and 2.69 for the Simulations 2021 and APAL Head 2019 datasets, which is $3.08\times$, and $2.34\times$ lower than without the fine-tuned TimeGPT model. This is because the fine-tuning optimizes the weights of TimeGPT and Chronos models to better align with the data distribution and patterns specific to cybersickness prediction, significantly enhancing their performance and practicality for real-world cybersickness prediction tasks. A similar phenomenon is also observed in the Chronos foundation model. On the other hand, we observe that using the pre-trained TimeGPT and Chronos foundation (without fine-tuning) does not perform well in cybersickness prediction tasks for both datasets. Thus, their generalized representations are inadequate for directly applied to the likelihood of cybersickness severity prediction. It is important to note that for the rest of the paper, we only consider the fine-tuned TimeGPT and Chronos model results in the cybersickness prediction task for both datasets. Furthermore, Table 2 summarizes the MAE, RMSE, and MAPE values for cybersickness prediction using ARIMA, LSTM, GRU, CNN-LSTM, NBEATS, NHITS, Transformer, Chronos and TimeGPT models for the simulations 2021 and APAL Head 2019 datasets. The cybersickness prediction using ARIMA, LSTM, GRU, CNN-LSTM, NBEATS, NHITS, Transformer, Chronos and TimeGPT models results in RMSE values of 1.89, 0.40, 0.45, 0.54, 0.38, 0.37, 0.34, 0.34 and 0.26 for the simulations 2021 and 14.45, 5.21, 5.65, 6.28, 4.91, 4.74, 3.59, 3.54, and 2.69 for the APAL Head 2019 datasets, respectively. From Table 2, we observe that the overall performance of the TimeGPT foundation model is better than the Chronos foundation

Table 1: The results of the TimeGPT and Chronos foundation models with and without fine-tuning (FT) in cybersickness prediction task for Simulations 2021 (Sim21) and APAL Head 2019 (APAL19) datasets. The notation X/Y represents the TimeGPT/Chronos models.

Metrics	Without FT (Sim21)	With FT (Sim21)	Without FT (APAL19)	With FT (APAL19)
MAE	0.47/0.81	0.18/0.21	4.67/6.64	2.08/2.91
RMSE	0.83/1.02	0.27/0.32	6.25/8.76	2.79/3.54
MAPE	0.62/0.87	0.21/0.24	5.02/7.68	2.41/3.03

Table 2: Performance of cybersickness prediction models using a 10-fold cross-validation technique and the model's total training time (MTTT) in minutes along with model's total inference (Inf) in seconds for the Simulations 2021 and APAL Head 2019 datasets. The notation X/Y represents the Simulations 2021/APAL Head 2019 dataset.

Models	MAE	RMSE	MAPE	MTTT (min)	Inf (sec)
ARIMA	1.20/10.32	1.89/14.45	1.06/8.45	189/212	0.11/0.18
LSTM	0.29/4.02	0.40/5.29	0.35/4.78	358/391	4.05/4.87
GRU	0.31/4.46	0.45/5.15	0.39/5.13	373/402	4.94/5.41
CNN-LSTM	0.37/5.11	0.54/5.88	0.48/6.05	413/428	4.75/4.98
NBEATS	0.27/3.79	0.38/4.91	0.32/4.02	396/434	4.83/5.16
NHITS	0.25/3.92	0.37/4.74	0.29/4.19	446/463	5.03/5.48
Transformer	0.21/3.03	0.34/3.59	0.25/3.14	682/703	5.48/6.07
Chronos	0.21/2.91	0.32/3.54	0.24/3.03	153/195	3.91/4.33
TimeGPT	0.18/2.08	0.26/2.69	0.21/2.41	125/147	3.69/3.95

model and other DL models in terms of MAE and MAPE values for the cybersickness prediction task for both simulations 2021 and APAL Head 2019 datasets. Overall, the TimeGPT model performs better than other models, even the Chronos foundation model, for both datasets. Furthermore, each model's total training time and inference time for simulations 2021 and APAL Head 2019 datasets are shown in Table 2. It is observed that the training time is significantly lower for the FSL-based fine-tuned TimeGPT and Chronos foundation models than other DL models (training from scratch) in the cybersickness prediction task for both datasets.

4 CONCLUSION

This paper proposed an innovative approach to cybersickness prediction using the pre-trained large foundation model. We validated our method on Simulations 2021 and APAL Head 2019 datasets, achieving superior accuracy and efficiency compared to traditional DL models. The fine-tuned TimeGPT model outperformed others with RMSE values of 0.26 and 2.69 for the Simulations 2021 and APAL Head 2019 datasets, which is $1.31\times$ and $1.39\times$ lower than the Transformer model, respectively.

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